UMANG VEKARIYA

Computer Science (AI & ML) Undergraduate | VIT Vellore

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Seeking AI Engineer and Software Engineering internships. Building computer-vision experiments, compilers, local agent evaluation tools, and dependable full-stack systems.

EDUCATION

Vellore Institute of Technology, Vellore | 2024 - Present
B.Tech, Computer Science (AI & ML) | Current CGPA: 9.21/10

FIITJEE Junior College | 2022 - 2024 | Grades 11-12: 92.5%
Narayana School | 2021 - 2022 | Grade 10: 9.6/10 GPA

TECHNICAL SKILLS

Languages: Python, C++, Java, C, JavaScript, TypeScript

AI & computer vision: TensorFlow, OpenCV, MediaPipe, scikit-learn, NumPy, SciPy

Web & APIs: React, Tailwind CSS, FastAPI, Node.js, Express.js, HTML / CSS

Data & infrastructure: PostgreSQL, MySQL, MongoDB, SQLite, Azure, AWS S3, Docker, Git

EXPERIENCE

Software Engineering Intern - Mahant Solutions

MAY - JUL 2026

Contributed to full-stack development, feature implementation, debugging, testing, and maintenance. Collaborated through version-control workflows and agile development practices.

Edge AI & Robotics Summer School - ACM India | IISc Bengaluru

JUL 2026

Selected participant in the ACM India Summer School at IISc. Explored edge AI, embedded machine learning, robotics, computer vision, and autonomous systems through expert-led sessions and hands-on activities.

PROJECTS

PolyGlot | Case study

Python | JavaScript | Typed IR | Custom bytecode VM | Pyodide / WebAssembly

- Built a lexer, parser, AST, semantic analysis, typed intermediate representation, bytecode format, and stack virtual machine.
- Created a shared CLI and browser IDE with compiler-stage inspection and reusable bytecode artifacts. Browser execution runs through Pyodide in a Web Worker.

WNAMS | Case study

React | TypeScript | FastAPI | SQLite | NumPy | SciPy | Swift CoreWLAN

- Calibrated floor plans and authenticated survey sessions using real laptop measurements.
- Five-sample RSSI collection, IDW/RBF coverage heatmaps, and channel-overlap analysis.

VIT Vault | Case study

Python | FastAPI | PostgreSQL | SQLite | Azure Blob / Queue Storage | Docker

- Co-developed an Azure-hosted workspace with versioned submissions, resources, private feedback, and collaboration.
- Built a campus LAN gateway for submission during internet outages, with local receipts and durable synchronization.

Deepfake Detection | Case study

Python | TensorFlow | MobileNetV2 | OpenCV | NumPy

- Used MobileNetV2 transfer learning for authentic-versus-manipulated video classification.
- Built an OpenCV preprocessing pipeline to extract video frames and prepare TensorFlow inputs.

RealityForge | Case study

TypeScript | MediaPipe | Web Workers | WebAssembly | HTML Canvas

- Used a planar desk calibration and foreground silhouettes to drive interactive simulations.
- Evaluated segmentation and tracking on six annotated synthetic clips containing 144 frames, including lighting changes, shadows, and hand occlusion.

PlayableReplay | Case study

Python | OpenCV | NumPy | JavaScript | HTML Canvas

- Implemented capture, object tracking, contact extraction, physics fitting, reconstruction, takeover, and verification.
- Fitted motion and collision parameters on the first 60% of observations. Evaluation uses five synthetic fixtures.

Intent Lab | Case study

Python | scikit-learn | TF-IDF | Logistic Regression | Qwen3-4B | Ollama

- Integrated Qwen-generated plans, separate permission checks, and execution receipts in a browser demo. The demo uses fictional, in-memory email tools.
- Collected 36 local Qwen development runs and 72 actual proposals, compared versioned models on frozen synthetic evaluations, and added a queue for independent label review.

ClubSphere | Case study

React | JavaScript | FastAPI | SQLAlchemy | MySQL | JWT

- Developed student, coordinator, and admin workflows for membership, booking, attendance, feedback, and reports.
- Designed nine normalized MySQL tables with keys, constraints, and indexes.

CERTIFICATIONS & COURSEWORK

Meta Front-End Developer Professional | Meta Advanced React | Meta Programming with JavaScript | Meta Introduction to Front-End Development | Meta React Basics | Meta HTML and CSS in Depth | Meta Coding Interview Preparation | Infosys Understanding Social Styles | Oracle Cloud Infrastructure AI Foundations | Infosys Basics of Python | Meta Version Control | Infosys Communicating to Succeed | Introduction to Generative AI Studio | Infosys Java Programming Fundamentals | Meta Principles of UX/UI Design | Meta Front-End Developer Capstone